A FINITE ETALE-COVERED SCHEME OVER A DENSE OPEN, REFINED BY AN ETALE COVER

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3925-3947, following the statement at 3910-3923.

Editorial note. No omissions; part (1) is the displayed special case of part (2).

Allowed standard prerequisites (editorial).

- Integral refined scheme over a dense open. Decent Spaces, Lemma decent-spaces-lemma-there-is-a-scheme-integral-over-refined (cross-chapter, cited at ~3929-3932). pi : Y -> X integral with Y a scheme, finite etale over a qcqs dense open U.
- Integral limits of finite presentation refinements. lemma-integral-limit-finite-and-finite-presentation-refined (same chapter, card L26287, cited at ~3934-3937). Y = lim Y_i with pi_i finite fp over X and pi^{-1}(U) -> pi_i^{-1}(U) an isomorphism.
- Limits that are schemes. lemma-limit-is-scheme (same chapter, cited at ~3938-3939). Y_i is a scheme for i large.
- Descent of opens. lemma-descend-opens (same chapter, cited at ~3940-3942). Affine opens W_{i,j} recovering W_j in the inverse image.
- Characterization of locally finite presentation spaces. proposition-characterize-locally-finite-presentation (same chapter, card L26289, cited at ~3943-3945). Lifts the factorizations W_j -> V to W_{i,j} -> U for i large.

Source reference. Original reference 0GUL: focused-seeds/geometry/S9-spaces-limits-there-is-a-scheme-finite-over-refined/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0827: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GUM: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3818–3818. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07SR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1435–1435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04AK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 694–694. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S such that $|X|$ has finitely many irreducible components.*

(1) *There exists a surjective finite morphism $f : Y \to X$ of finite presentation where Y is a scheme such that f is finite étale over a quasi-compact dense open $U \subset X$,*
(2) *given a surjective étale morphism $V \to X$ we may choose $Y \to X$ such that for every $y \in Y$ there is an open neighbourhood $W \subset Y$ such that $W \to X$ factors through V.*

Proof. Part (1) is the special case of (2) with $V = X$.

Proof of (2). Let $\pi : Y \to X$ be as in Decent Spaces, Lemma 0GUL and let $U \subset X$ be a quasi-compact dense open such that $\pi^{-1}(U) \to U$ is finite étale. Choose a finite affine open covering $Y = \bigcup W_j$ such that $W_j \to X$ factors through V. We can write $Y = \lim Y_i$ with $\pi_i : Y_i \to X$ finite and of finite presentation such that $\pi^{-1}(U) \to \pi_i^{-1}(U)$ is an isomorphism, see Lemma 0GUM. For large enough i the algebraic space Y_i is a scheme, see Lemma 07SR. For large enough i we can find affine opens $W_{i,j} \subset Y_i$ whose inverse image in Y recovers W_j, see Lemma 0827. For even larger i the morphisms $W_j \to V$ over X come from morphisms $W_{i,j} \to U$ over X, see Proposition 04AK. This finishes the proof. □